

$$1) \int \frac{(2x^4 + 7)(x^3 + 4)}{2\sqrt{x}} dx$$

$$(2x^4 + 7)(x^3 + 4)$$

$$2x^4 \cdot x^3 = 2x^7$$

$$2x^4 \cdot 4 = 8x^4$$

$$7 \cdot x^3 = 7x^3$$

$$7 \cdot 4 = 28$$

$$(2x^4 + 7)(x^3 + 4) = 2x^7 + 8x^4 + 7x^3 + 28$$

$$\int \frac{2x^7 + 8x^4 + 7x^3 + 28}{2\sqrt{x}} dx$$

$$\int \left(\frac{2x^7}{2\sqrt{x}} + \frac{8x^4}{2\sqrt{x}} + \frac{7x^3}{2\sqrt{x}} + \frac{28}{2\sqrt{x}} \right) dx$$

$$\frac{2x^7}{2\sqrt{x}} = x^{7-\frac{1}{2}} = x^{\frac{13}{2}}$$

$$\frac{8x^4}{2\sqrt{x}} = 4x^{4-\frac{1}{2}} = 4x^{\frac{7}{2}}$$

$$\frac{7x^3}{2\sqrt{x}} = \frac{7}{2}x^{3-\frac{1}{2}} = \frac{7}{2}x^{\frac{5}{2}}$$

$$\frac{28}{2\sqrt{x}} = 14x^{-\frac{1}{2}}$$

$$\int \left(x^{\frac{13}{2}} + 4x^{\frac{7}{2}} + \frac{7}{2}x^{\frac{5}{2}} + 14x^{-\frac{1}{2}} \right) dx$$

INTEGRALS

$$\int x^{\frac{13}{2}} dx = \frac{x^{\frac{15}{2}}}{\frac{15}{2}} = \frac{2}{15} x^{\frac{15}{2}}$$

$$\int 4x^{\frac{7}{2}} dx = 4 \cdot \frac{x^{\frac{9}{2}}}{\frac{9}{2}} = \frac{8}{9} x^{\frac{9}{2}}$$

$$\int \frac{7}{2} x^{\frac{5}{2}} dx = \frac{7}{2} \cdot \frac{x^{\frac{7}{2}}}{\frac{7}{2}} = x^{\frac{7}{2}}$$

$$\int 14x^{-\frac{1}{2}} dx = 14 \cdot \frac{x^{\frac{1}{2}}}{\frac{1}{2}} = 28x^{\frac{1}{2}}$$

$$\int \frac{(2x^4 + 7)(x^3 + 4)}{2\sqrt{x}} dx = \frac{2}{15} x^{\frac{15}{2}} + \frac{8}{9} x^{\frac{9}{2}} + x^{\frac{7}{2}} + 28x^{\frac{1}{2}} + C$$